



JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
KATOL ROAD, NAGPUR
Affiliated to DBATU, RTMNU & MSBTE, Mumbai
Website: www.jdcoem.ac.in E-mail: info@jdcoem.ac.in
An Autonomous Institute, with NAAC "A" Grade
Department of Computer Science & Engineering
"A place to learn; A chance to grow"
Session: 2025-26



MID SEMESTER EXAMINATION-01

Course: B. Tech. in Computer Science & Engineering

Year/Semester: 3rd/ 5th Sem

Subject: Machine Learning

Subject Code: CS5T002

Max Marks:20

Date: 05/08/2025

Duration:- 1 Hr.

Instructions to the Students:

1. All the questions carry marks as indicate.
2. Assume suitable data whenever necessary.
3. Illustrate your answer whenever necessary with the help of neat sketches.

Students should be able to:

1. Understand the basics of machine learning and different types of learning methods.
2. Apply concept learning and clustering algorithms to solve learning problems.
3. Analyze Bayesian learning approaches and learning theories.
4. Design predictive models using regression, kernel methods, and ensemble techniques.
5. Evaluate and compare machine learning algorithms using decision trees and hypothesis testing.

SECTION A

1. All Questions are compulsory.
2. Answer in one word/sentence.

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Define the Machine Learning.	L1/CO1	1
Q.2	Tell about the Clustering.	L1/CO2	1
Q.3	Draw the diagram showing the steps involved in designing a learning system.	L1/CO1	1
Q.4	List the Issues in Machine Learning.	L1/CO1	1

SECTION B

1. All Questions are compulsory.
2. Solve answer in brief (two, three Sentence)

Q.5	Describe the Advantages and Disadvantages of Unsupervised Machine Learning.	L2/CO1	2
Q.6	Discuss the Applications of Machine Learning.	L2/CO1	2
Q.7	Illustrate the Well Posed Learning Problems with Examples.	L3/CO1	2
Q.8	State the different Steps involved in the K-mean Algorithm.	L1/CO2	2

SECTION C

1. Solve any two questions.

Q.9	Describe the Life Cycle of Machine Learning.	L2/CO1	4
Q.10	Explain Hierarchical Agglomerative Clustering (HAC) and illustrate its workflow with a diagram.	L2/CO2	4
Q.11	Illustrate the different type of Machine Learning with Example.	L3/CO1	4